WASP-50b

R-0878

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-50_171115` · created 2026-07-29T00:21:19+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458072.8675 +/- 0.0026 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1469 +/- 0.0099
Transit depth (Rp/Rs)^2	2.16 +/- 0.29 [%]
Orbital Inclination (inc)	86.01 +/- 1.32
Airmass coefficient 1 (a1)	0.9898 +/- 0.0084
Airmass coefficient 2 (a2)	0.004 +/- 0.0026
Scatter in the residuals of the lightcurve fit is	0.85 %
Best Comparison Star	#3 - [240, 265]
Optimal Aperture	3.84
Optimal Annulus	8.33
Transit Duration (day)	0.0831 +/- 0.0079

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1469 ± 0.0099 vs 0.1404 (deviation 0.66σ)
Duration fit vs published	0.0831 d vs 0.0754 d (deviation 0.97σ)
Mid-time O–C	7.7 min
Depth SNR	14.8
Residual scatter	0.85 %

Light curve

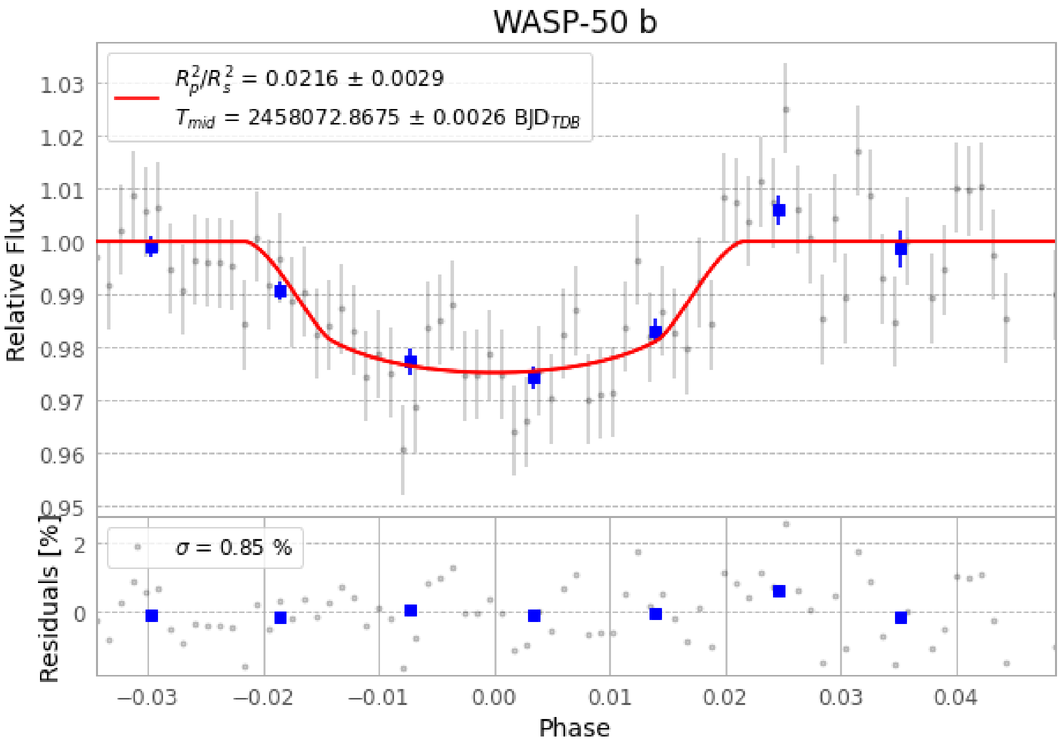

Figure 1 — FinalLightCurve_WASP-50 b_2017-11-15.png (SHA-256 6ffd6c1b19a44583…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [252, 287], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 f4b80a45ecf9b769…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

79 FITS frames (52 MB), `WASP-50171115070312.FITS` → `WASP-50171115105712.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-171115.log (5978b6b43409…), FinalLightCurve_WASP-50 b_2017-11-15.png (6ffd6c1b19a4…), FinalLightCurve_WASP-50 b_2017-11-15.csv (e36e859451d2…)

Compute resources

Wall time 175.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-29 00:21Z.